

The broad $\mathfrak{G}$ -base characterization for $L(X)$ is answered by arXiv:1606.01967

Literature-status note for arXiv:1604.02555

17 August 2026

Status. Explicit later-paper answer to broad Problem 1.1. This does not settle the separate question of removing the Ascoli hypothesis from the source paper’s compact-resolution reformulation, nor its broad Problem 1.3. This note records a literature identification, not a new theorem.

1 The source problem

For a Tychonoff space X , let $L(X)$ denote its free locally convex space. Gabrielyan and Kačol ask in Problem 1.1 on PDF page 2 of arXiv:1604.02555v2 [1]:

For which spaces X does the free locally convex space $L(X)$ have a $\mathfrak{G}$ -base?

Their terminology means a neighborhood base at zero indexed monotonically by $\mathbb{N}^{\mathbb{N}}$. The problem is reproduced from Question 4.15 of an earlier paper by Gabrielyan, Kačol, and Leiderman.

The source proves a useful Ascoli-space reformulation: if X is Dieudonné complete and Ascoli, then $L(X)$ has a $\mathfrak{G}$ -base exactly when $C_k(X)$ admits a compact resolution swallowing compact subsets [1, Theorem 1.2].

2 The explicit complete characterization

Banach and Leiderman state in the introduction of arXiv:1606.01967v2 that their Theorem 8.1 “resolves completely” Question 4.15 and repeat the question in the same sentence [2, PDF pp. 2–3]. Theorem 8.1(1), on PDF page 19, gives the promised characterization for every Tychonoff space X :

$$\begin{array}{l} L(X) \text{ has a local } \mathfrak{G}\text{-base} \quad \Longleftrightarrow \\ \mathcal{U}(X) \text{ has a } \mathfrak{G}\text{-base and } C(X) \text{ is } \mathbb{N}^{\mathbb{N}}\text{-dominated.} \end{array}$$

Here $\mathcal{U}(X)$ is the finest compatible uniformity, $C(X)$ is ordered pointwise, and $\mathbb{N}^{\mathbb{N}}$ -dominated means that there is a monotone family $(f_\alpha)_{\alpha \in \mathbb{N}^{\mathbb{N}}} \subseteq C(X)$ cofinal for the pointwise order. The theorem also gives the equivalent condition that $C(X)$ is $\mathbb{N}^{\mathbb{N}}$ -dominated as a subposet of $\mathbb{R}^X$, and equivalence with the free topological vector space $V(X)$ having a local $\mathfrak{G}$ -base.

Since a $\mathfrak{G}$ -base for a topological vector space is, by definition, such a local base at zero, this is a necessary-and-sufficient answer to the source’s broad Problem 1.1, with no Ascoli, separability, or metrizability hypothesis.

3 Scope that remains open

This identification must not be overread. On PDF page 8, the source paper explicitly says that it does not know whether the Ascoli assumption in its Theorem 1.2 is essential. The later Theorem 8.1 does not remove that assumption from the particular equivalence involving a compact resolution in $C_k(X)$: its item (7), relating $L(X)$ to a local $\mathfrak{G}$ -base of $C_k(C_k(X))$, still assumes X is Ascoli. Thus the broad intrinsic characterization is complete, while this sharper reformulation question is not claimed here.

Likewise, Problem 1.3 of the source asks more broadly which spaces X make $C_k(X)$ admit a compact resolution swallowing compact subsets. The cited theorem is not a characterization of that class, so this packet does not dispose of Problem 1.3.

References

- [1] S. Gabrielyan and J. Kaşol, *Free locally convex spaces with a small base*, arXiv:1604.02555v2 (2016).
- [2] T. Banach and A. Leiderman, *$\mathfrak{G}$ -bases in free (locally convex) topological vector spaces*, arXiv:1606.01967v2 (2016).